

Profitable Cheap Talk and Zero-Retaliation Rent: Why Trust-Restoring Signals Fail Before Common Knowledge

Anonymous Author(s)

Submission #XXX, USA

Abstract

We study a signaling environment in which the messages by which posteriors would become common knowledge are themselves profitable to distort. A sender classifies a payoff-relevant object that is not a strategic player and therefore cannot impose refusal, exit, sanction, litigation, or retaliation costs on false classification. Anthropomorphic AI marketing is the leading application. At this zero-retaliation boundary, trust-restoring messages are not merely cheap talk; they are profitable cheap talk. We show that pooling survives neologism-proofness: every trust-restoring message available to the high-governance type is profitably shadowable by the low-governance type. Voluntary non-arbitrage is therefore not incentive compatible. Separation requires externally imposed signal costs satisfying single crossing, such as audit trails, sanctionable inconsistency, and auditable records. A dynamic extension embeds the baseline game in an endogenous trust-capital law of motion and shows that cheap anthropomorphic signals, user habituation, and delayed harms do not by themselves break pooling: when the semantic premium and engagement reinforcement offset bounded asynchronous shocks, the pooling interval is almost surely forward-invariant.

Keywords: Signaling games, profitable cheap talk, zero-retaliation rent, endogenous trust, dynamic pooling, neologism-proofness, costly signaling, auditable records

1. Introduction: Before Common Knowledge

Aumann showed that rational agents with a common prior cannot agree to disagree once their posteriors are common knowledge (Aumann, 1976). But how do posteriors become common knowledge when the messages that would reveal them are themselves profitable to distort? They often do not. In profitable cheap-talk environments, trust-restoring messages are rationally mimicked by the very types they are meant to exclude.

Anthropomorphic AI marketing is a limit case because the classified object is payoff-relevant but cannot retaliate. A firm can sell the system as a companion, tutor, or collaborative agent in the consumption channel while describing the same system as a mere tool in liability-facing channels. The resulting surplus is *zero-retaliation rent*: a premium earned on a classification boundary that carries no signal cost from the classified object. Standard cheap talk assumes costless messages. This paper studies a more pathological case: messages whose production cost is zero but whose equilibrium payoff is positive. The signal is not merely free to send; it is profitable to send.

The failure studied here occurs before Aumann agreement can begin. The problem is not that rational agents continue to disagree after posteriors become common knowledge. It is that the messages that would make posteriors common knowledge are themselves rent-generating. When a low-governance sender can profit from sending the same trust-restoring message as a high-governance sender, and when the classified object cannot impose retaliation costs on that message, truthful restraint is not an equilibrium strategy. The result is profitable cheap talk.

The contribution is five concrete claims:

1. profitable cheap talk at the zero-retaliation boundary: the anthropomorphic signal has negative net cost for both governance types because the classified object cannot retaliate;
2. neologism-proof shadowing: every trust-restoring message available to the high-governance type is profitably shadowable, so voluntary non-arbitrage

	Comply	Resist
Objectify	$(L + Sec, -C_{sub})$	$(L + Sec - C_{enf}, -C_{sub} - C_{res})$
Recognize	$(Eth - Sec, +V)$	$(Eth - Sec, +V)$

Table 1: The inadequate dyadic model. The static Nash matrix is retained only as a foil: it omits private types, public messages, posterior beliefs, and the regulatory channel through which ontological claims become governable.

is not incentive compatible;

3. cost-imposed common-knowledge formation: separation requires externally imposed signal costs satisfying single crossing;
4. pricing the ontological spread: sanctionable inconsistency, audit requirements, and auditable records each close a different term of the low-type arbitrage spread, and their necessity and sufficiency conditions are stated as a unified pricing proposition;
5. dynamic extension: endogenous trust dynamics can lock in pooling, history-dependent trust-restoring messages remain shadowable, and dynamic separation requires an externally priced single-crossing wedge.

The paper organizes the mathematics into a baseline proposition and dynamic extensions of the same signaling problem. Proposition 1 establishes pooling on profitable cheap talk as a Perfect Bayesian Equilibrium. Theorem 1 proves that the pooling equilibrium is neologism-proof at the zero-retaliation boundary due to profitable shadowing. Theorem 2 proves cost-imposed common-knowledge formation under single crossing. Section 7 then embeds the same incentives in a trust-capital state equation and proves dynamic pooling invariance, dynamic shadowing, and externally priced dynamic separation: history and delayed harms do not cure pooling unless some signal cost, audit price, or sanctionable inconsistency changes the Bellman value wedge.

2. Model: Payoff-Relevant Non-Player as the Negative-Cost Signal Environment

The standard starting point for ontological conflict is a dyadic observer–subject model; Table 1 compresses that foil into a simultaneous Nash matrix that identifies a real temptation but omits types, messages, beliefs, and the regulatory channel. The more basic problem is the player set. In anthropomorphic AI markets, the system is the object around which strategic claims are made, but it is not a strategic player: it has no channel through which to impose costs on those claims. At the zero-retaliation limit, this cost class is absent. What the stylized governance setting therefore requires is a three-sided structure: a Firm (F) that produces ontological claims, a User (U) that consumes them, and a Regulator (R) whose posterior over firm type determines whether the claims become governable. This section fixes the player intuition that the formal apparatus of Section 3 then makes precise.

2.1. The Firm Side

A firm deploying a model has private knowledge of two parameters: its internal *governance type* θ_F (the degree of investment in evaluations, red-teaming, and post-deployment monitoring) and the system’s *opacity* θ_S (the extent to which capability claims can be checked from outside). Both are unobservable to users and to regulators. What is observable is the firm’s *message*: a pair of public artefacts—marketing copy and policy/liability documentation—which together constitute the firm’s signal m_F .

The signal admits two stable poles. *Anthropomorphic marketing* attributes agency, judgment, or care to the system in ways that warrant a relational reading. *Deflationary policy* disclaims exactly those attributes in ways that warrant a tool reading. The firm enjoys an *ontological premium*: a utility surplus that accrues when the system is priced as agent in the consumption channel (driving willingness-to-pay and engagement) while being priced as tool in the liability channel (limiting redress). The premium is rent earned on a category boundary

the firm itself does not have to defend. This is not a claim about firm pathology; it is a claim about the payoffs any firm faces under the present signal-cost structure. A firm with $\theta_F = \mathbf{H}$ that elected deflationary marketing would forfeit revenue to a competitor who did not, absent an external instrument that taxes the asymmetry.

2.2. The User Side

Users carry a private *vulnerability type* θ_U that aggregates affective need, isolation, and the marginal value of a high-availability interlocutor. Their action $m_U \in \{\text{invest}, \text{detach}\}$ is the decision to extend or withhold investment: to act on the anthropomorphic signal or to discount it. The signal is observable, but the type is not. Users cannot inspect training data, evaluations, or internal incident reports; their evidence is the firm’s message and the system’s behaviour, both of which are jointly designed. This information asymmetry compounds because the harms of mis-investment are *asynchronous*: the cost of acting on the anthropomorphic signal falls due in episodes (model deprecations, persona changes, refusals at moments of acute need) that may be separated from the original investment by months. A risk-neutral user under cheap-talk signals will rationally invest, because the immediate utility of invest exceeds the discounted expected cost of liquidation. Section 8.1 illustrates the consumer-side informational asymmetry with a concrete marketing–documentation example.

2.3. The Regulator Side and the Public Channel

A regulator R chooses among *inspect*, *sanction*, and *abstain*, conditional on a posterior over θ_F and θ_S formed from the firm’s public messages and from an exogenous public signal z . The regulator’s private type θ_R is *bandwidth*: the capacity, technical and institutional, to discriminate genuine capability claims from cheap-talk ones. A high-bandwidth regulator can read whitepapers, commission audits, and detect inconsistency across the firm’s two channels; a low-bandwidth regulator must rely on the channel z .

The public channel z is the aggregate output of journalism, academic critique, civil-society advocacy, and incident reporting. We treat it as a noisy estimator of θ_F : it shifts R 's posterior but does not deliver verdicts. Folding publics into R this way is deliberate. Publics rarely have standing to sanction; they have standing to be *heard*, and the channel through which they are heard is the regulator's information environment. Treating intellectuals and affected communities as autonomous players would over-parameterise the model without changing its qualitative behaviour: in every equilibrium we examine, what publics do is move R 's posterior; what changes the firm's behaviour is what R does next.

The three-sided structure thus admits a single arbitrage logic. The firm earns rent on the spread between two ontological claims it makes about the same artefact. The user supplies the demand that makes the spread profitable. The regulator, in the absence of a verification mechanism that ties the two claims together, lacks an off-path belief stringent enough to deter the spread. The remainder of the paper formalises this structure (Section 3), shows why trust-restoring messages fail (Section 4), explains why profitable cheap talk persists (Section 6), and shows how its signal-cost structure may be redesigned (Section 8).

Informal preview of the equilibrium analysis. A firm's governance quality is its private information. Under current conditions, describing an AI system as a companion or agent has zero production cost and positive continuation value. Because the signal is rent-generating for both governance types, it carries no type information: users cannot distinguish careful firms from careless ones, and regulators cannot justify the cost of inspection. This is the pooling equilibrium. Separation requires pricing the signal. If anthropomorphic marketing is valid only when paired with an audit trail—evaluation records, capability logs, incident reports—and if the cost of producing that trail is lower for firms that actually invested in governance (the single-crossing condition), then a threshold audit intensity exists above which low-governance firms will not imitate and be-

low which high-governance firms can still afford to signal. A further asymmetry governs the size of the arbitrage premium itself. In prior recognition games, the classified subject retained a non-zero capacity to impose costs on the arbitrageur, which discounted the premium. The AI case is the limit where the subject’s retaliation capacity is zero: the system has no independent channel through which to impose costs on the firm, leaving the ontological premium unconstrained. This structural boundary is formalized in Section 3.

3. Baseline Pooling: Rational Arbitrage, Not Moral Failure

This section formalizes the three-sided market dynamics as a sequential signaling game under incomplete information.

3.1. Model

Players. The strategic players are a Firm (F), a User (U), and a Regulator (R). The AI system is not a player but an object of classification. Publics enter as a noisy exogenous signal $z \in Z$.

Type Spaces. Nature draws private types from a common prior π over $\Theta = \Theta_F \times \Theta_U \times \Theta_R \times \Theta_S$. The firm’s governance type is $\theta_F \in \{\mathbf{H}, \mathbf{L}\}$. The user’s vulnerability type is $\theta_U \in \{\mathbf{V}, \mathbf{N}\}$. The regulator’s bandwidth type is $\theta_R \in \{\mathbf{B}, \bar{\mathbf{B}}\}$. The system-opacity parameter θ_S is private to F .

Baseline Message Spaces. In the unpriced baseline, the firm chooses $m_F \in \{a, p\}$, where a is anthropomorphic marketing and p is policy-deflationary positioning. The audit-augmented message space is introduced only in Theorem 2, because audit intensity is an externally imposed signal cost rather than part of the cheap-talk baseline. The user chooses $m_U \in \{i, d\}$ (invest or detach). The regulator chooses $m_R \in \{x, s, o\}$ (inspect, sanction, or abstain).

Timing. 1. F observes (θ_F, θ_S) and sends m_F . 2. U observes m_F and θ_U , then sends m_U . 3. R observes (m_F, m_U, z) and θ_R , then chooses m_R .

Histories. The user’s public history is $h_U = (m_F)$, and the regulator’s is $h_R = (m_F, m_U, z)$.

Strategies. A firm strategy $\sigma_F : \Theta_F \times \Theta_S \rightarrow \Delta(\{a, p\})$. A user strategy $\sigma_U : \{a, p\} \times \Theta_U \rightarrow \Delta(\{i, d\})$. A regulator strategy $\sigma_R : \{a, p\} \times \{i, d\} \times Z \times \Theta_R \rightarrow \Delta(\{x, s, o\})$.

Beliefs. The user forms posterior $\mu_U(\theta_F, \theta_S \mid h_U)$. The regulator forms posterior $\mu_R(\theta_F, \theta_S \mid h_R)$.

Payoff Functions. We consolidate the payoffs for Firm, User, and Regulator into the following block:

$$\begin{aligned} u_F(m_F, m_U, m_R) &= \begin{cases} \Delta_F^S - c_{\theta_F}(m_F) - P(m_R) & \text{if } m_F = a, m_U = i \\ -c_{\theta_F}(m_F) - P(m_R) & \text{otherwise} \end{cases} \\ u_U(m_F, m_U, \theta_F) &= \begin{cases} B_{\theta_U} & \text{if } m_U = i, \theta_F = H \\ B_{\theta_U} - H_{\theta_U} & \text{if } m_U = i, \theta_F = L \\ 0 & \text{if } m_U = d \end{cases} \\ u_R(m_F, m_U, m_R, \theta_F) &= \begin{cases} -K_{\theta_R} + D_R & \text{if } m_R \in \{x, s\}, \theta_F = L \\ -K_{\theta_R} & \text{if } m_R \in \{x, s\}, \theta_F = H \\ 0 & \text{if } m_R = o \end{cases} \end{aligned}$$

where Δ_F^S is the ontological premium, $c_{\theta_F}(m_F)$ is the production or verification cost of the signal, $P(m_R)$ is the penalty if R sanctions, B_{θ_U} is the immediate benefit of investment, H_{θ_U} is the future harm of false investment, K_{θ_R} is R 's inspection cost, and D_R is R 's payoff from catching a low-governance firm. The object that matters for the paper is the signal's *net* cost:

$$\text{NetCost}_{\theta}(a) := c_{\theta}(a) - \Delta_F^S.$$

At the zero-retaliation boundary, we have $\Delta_F^S = \Delta_F$, so Assumption 1 implies $\text{NetCost}_{\theta}(a) = -\Delta_F < 0$ for both governance types. The message is therefore rent-generating.

Equilibrium Concept. We employ Perfect Bayesian Equilibrium (PBE) (Kreps and Wilson, 1982) supported by Farrell's neologism-proofness (Farrell, 1993).

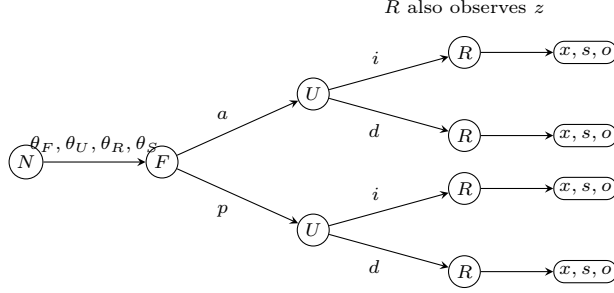


Figure 1: Extensive-form skeleton of the signaling game. Nature draws private types; F signals first, U responds, and R acts after observing both public messages and the noisy public signal z .

Assumptions.

Assumption 1 (Cheap-talk costs). $c_H(a) = c_H(p) = c_L(a) = c_L(p) = 0$.

Assumption 2 (Pooling region). $\Delta_F > 0$, $B_{\theta_U} - \pi(L)H_{\theta_U} > 0$, $K_{\theta_R} > \pi(L)D_R$.

3.2. Equilibrium Results

Proposition 1 (Profitable cheap-talk pooling). *Under Assumptions 1 and 2, pooling on anthropomorphic marketing is a Perfect Bayesian Equilibrium. At $\rho_S = 0$, the on-path anthropomorphic message has negative net signal cost for both governance types.*

Proof. Consider the strategy profile where both firm types send a , both user types invest after a , and the regulator abstains. Off path, users detach after p and R abstains. Sequential rationality holds: a firm deviating to p loses Δ_F^S because users detach, without reducing penalties because R already abstains. Users find investment optimal under the prior by Assumption 2. Regulators abstain under the prior by Assumption 2. Bayes consistency is satisfied because posteriors equal priors on path, and off-path beliefs are fixed at the prior. $\square$

Corollary 1 (Voluntary non-arbitrage is not incentive compatible). *For type θ in the pooling region, let $\mathbb{E}[P(m)]$ denote the expected penalty following message*

$m \in \{a, p\}$. If

$$\Delta_F^S > c_\theta(a) - c_\theta(p) + \mathbb{E}[P(a)] - \mathbb{E}[P(p)],$$

then choosing a strictly dominates voluntary deflationary positioning p for type θ . Hence voluntary non-arbitrage is not incentive compatible.

Proof. In the pooling region, receiver beliefs and continuation behaviour are fixed as in Proposition 1. The payoff difference for type θ between choosing a and choosing p is

$$u_\theta(a) - u_\theta(p) = \Delta_F^S - c_\theta(a) + c_\theta(p) - \mathbb{E}[P(a)] + \mathbb{E}[P(p)].$$

The displayed hypothesis is exactly the condition that this difference is strictly positive. Therefore a strictly dominates p , so unilateral voluntary refusal of the ontological premium is not a best response. $\square$

4. Shadowing and Neologism-Proofness: Why Trust-Restoring Signals Fail

Lemma 1 (Equal continuation premium under zero-cost mimicry). *Under Assumption 1, the firm's payoff from any trust-restoring continuation c that induces user investment and regulatory abstention is type-independent:*

$$u_H(c) = u_L(c) = \Delta_F^S.$$

Therefore $u_H(c) > \bar{u}_H$ if and only if $u_L(c) > \bar{u}_L$.

Proof. Under continuation c , the user invests ($m_U = i$) and the regulator abstains ($m_R = o$), so $P(m_R) = 0$. The firm's payoff is $\Delta_F^S - c_{\theta_F}(a) - P(m_R) = \Delta_F^S - c_{\theta_F}(a)$. Assumption 1 sets $c_H(a) = c_L(a) = 0$, so both types receive Δ_F^S . Since pooling-equilibrium payoffs $\bar{u}_H$ and $\bar{u}_L$ are also type-independent under Assumption 1 (both equal Δ_F^S from Proposition 1), the strict-gain comparison reduces to $\Delta_F^S > \bar{u}_\theta$ for both θ simultaneously. $\square$

Theorem 1 (Profitable shadowing at zero retaliation). *When $\rho_S = 0$, the pooling equilibrium of Proposition 1 is neologism-proof against every nontrivial*

trust-restoring type claim: whenever a continuation is strictly profitable for the high-governance type, it is also shadowable by the low-governance type.

Proof. A neologism K is credible only if the types in K strictly prefer the induced continuation c , while types outside K do not. Take the only nontrivial trust-restoring claim, $K = \{H\}$. At $\rho_S = 0$, the subject cannot impose any independent cost and $\Delta_F^S = \Delta_F$. The continuation c that restores trust induces user investment and regulatory abstention. By Lemma 1, the firm's payoff under c is type-independent: $u_H(c) = u_L(c) = \Delta_F^S$. Therefore

$$u_H(c) > \bar{u}_H \implies u_L(c) > \bar{u}_L,$$

so the deviation cannot exclude L. The trust-restoring neologism is not credible in Farrell's sense. $\square$

The point is not that the low-governance type is merely able to mimic. Whenever the continuation premium exceeds expected penalty, mimicry is itself the low-governance type's best response. Trust-restoring neologisms therefore fail before they can make governance quality common knowledge: the message that would form common knowledge is profitably shadowable.

Corollary 2 (Zero-retaliation boundary). *Let the effective premium be $\Delta_F^S = \Delta_F - \rho_S C_S$, where $\rho_S \in [0, 1]$ is the subject's retaliation capacity and C_S is the cost imposed through resistance, exit, litigation, or sanction. The boundary $\rho_S = 0$ removes this constraint class and leaves the premium unconstrained by subject retaliation.*

Proof. When $\rho_S > 0$, the subject can impose expected cost $\rho_S C_S$, shrinking the pooling region. When $\rho_S = 0$, $\Delta_F^S = \Delta_F$, so the off-equilibrium punishment path disappears. $\square$

Remark 1 (Agentic systems and ρ_S). *The model distinguishes causal capacity from payoff-imposing capacity. Tool use is not retaliation. The parameter ρ_S does not measure whether an AI system can send emails, post messages, call tools, or coordinate activity through firm-provided affordances. It measures*

whether the classified object can impose an independent cost on the sender's classification of it. Agentic AI may increase public visibility, detection probability, or regulator bandwidth, thereby changing z , K_R , D_R , or expected penalties. But unless the system has a protected and non-waivable channel of refusal, exit, sanction, litigation, or cost-imposition, its subject-retaliation capacity remains zero. A future regime that grants such a channel is covered by the comparative statics $\Delta_F^S = \Delta_F - \rho_S C_S$; the zero-retaliation case studied here is the limit in which behavioral agency does not yet amount to strategic standing.

4.1. Robustness of Off-Path Beliefs

A standard critique of pooling equilibria is their reliance on particular off-path beliefs. In Theorem 1, off-path beliefs are fixed at the prior, and the user detaches after observing a deviation to p . One might argue that the equilibrium would collapse under stronger refinement concepts. We show that any off-path belief assigning credibility to a high-governance deviation is defeated by L-type mimicry.

If a receiver assigns a high posterior to H following an out-of-equilibrium message m' , the message induces user investment. But because signaling costs are zero (Assumption 1), the L-type firm can trivially imitate m' . Since Δ_F^S is captured uniformly, the L-type strict deviation payoff strictly exceeds its pooling payoff whenever the H-type deviation payoff does.

Consequently, standard type-exclusion refinements like the Intuitive Criterion (Cho and Kreps, 1987) and Divinity (Banks and Sobel, 1987) do not select the desired separating deviation. The Intuitive Criterion eliminates types for whom the deviation is equilibrium-dominated regardless of the receiver's best response. Here, neither type is equilibrium-dominated because both types strictly benefit if the receiver invests. Farrell's neologism-proofness (Farrell, 1993) therefore becomes the mathematically appropriate refinement: it asks whether a message with a literal meaning can be believed. As Theorem 1 shows, the zero-retaliation limit ensures perfect shadowing, meaning no such belief can be sustained in equilibrium. The pathology is not that off-path beliefs are too

weak, but that any trust-restoring belief is immediately devoured by shadowing.

5. Cost-Imposed Separation: How Common Knowledge Becomes Possible

Separation is not moral communication. It is engineered common-knowledge formation through externally imposed signal costs. In the baseline, a is profitable cheap talk because $\text{NetCost}_\theta(a) < 0$ for both governance types. An audit requirement changes the message space and asks whether the trust-restoring signal can be priced so that the high-governance type can still send it while the low-governance type cannot profitably shadow it.

Theorem 2 (Cost-imposed common-knowledge formation). *Consider the audit-augmented message space*

$$M_F^e = \{p, a_0\} \cup \{\hat{a}(e) : e \geq 0\},$$

where p is deflationary positioning, a_0 is an unaudited anthropomorphic claim, and $\hat{a}(e)$ is an anthropomorphic claim accompanied by an audit trail of intensity e . Let G denote the gross continuation gain to the firm from having $\hat{a}(e)$ accepted as credible, before paying the audit cost. Equivalently, G is the trust-restoration premium: in the benchmark payoff normalization above $G = \Delta_F^S$, while if audit acceptance also changes the regulator's continuation from inspection or sanction to abstention, G includes the avoided expected penalty. Under an audit-trail signal cost satisfying the single-crossing condition

$$c_H(e, \theta_S) \leq G \leq c_L(e, \theta_S),$$

a separating Perfect Bayesian Equilibrium exists.

Proof. Let $c_\theta(e, \theta_S) = k_\theta e + \theta_S$, with $0 < k_H < k_L$. For an audit intensity

$$e \in \left[\frac{G - \theta_S}{k_L}, \frac{G - \theta_S}{k_H} \right],$$

the high-governance type prefers the audited anthropomorphic signal $\hat{a}(e)$, netting $G - c_H(e, \theta_S) \geq 0$, while the low-governance type does not mimic because

$G - c_L(e, \theta_S) \leq 0$. Beliefs assign **H** to $\hat{a}(e)$ and **L** to the deflationary signal. The unaudited claim a_0 receives the same posterior treatment as profitable cheap talk and therefore does not produce the trust-restoring continuation G . Given these beliefs, user investment and regulatory abstention after $\hat{a}(e)$, and detachment or inspection after non-audited signals, are sequentially rational in the separating region. $\square$

The theorem is the Spence step in the paper. A trust-restoring signal becomes credible only when the external cost schedule reverses the negative net cost of the baseline signal and satisfies single crossing. The high type must face

$$\text{NetCost}_H(\hat{a}(e)) = c_H(e, \theta_S) - G \leq 0,$$

while the low type must face

$$\text{NetCost}_L(\hat{a}(e)) = c_L(e, \theta_S) - G \geq 0.$$

Only then can the audited signal support common-knowledge formation rather than another shadowable message.

Theorem 2 gives the local separation condition. The mechanism consequences—sanctionable inconsistency, audit requirements, and auditable records—are stated as Corollaries 6–8 in Section 8 after introducing a common notation for the low-type arbitrage spread. That notation is useful because the three interventions operate on different terms of the same net-payoff inequality.

6. Why Profitable Cheap Talk Does Not Self-Correct

Theorem 2 identifies the cost condition that restores separation. This section explains why profitable cheap talk persists when that condition is absent. Each persistence mechanism is matched to a parameter of the model so that the proposals in Section 8 can be read as targeted cost changes rather than as welfare exhortations. They preserve negative net signal cost or prevent the penalty term from exceeding the premium.

Opacity. The system-opacity parameter θ_S is private to the firm. Capability claims about a deployed model cannot be falsified by reading model outputs alone, because the same outputs are compatible with very different training procedures, evaluation regimes, and post-deployment monitoring. Opacity is the precondition for the firm’s premium: where outsiders could costlessly verify θ_F , the spread between anthropomorphic and deflationary positioning would collapse. This is the lemons logic of Akerlof (1970): under one-sided private information, the market clears at a price that reflects the worst credible quality, except that here the relevant “quality” is a governance disposition rather than a physical attribute.

Verification bandwidth. The third inequality of the pooling region, $K_{\theta_R} > \pi(L)D_R$, encodes the regulator’s inspection cost relative to expected gain. A low-bandwidth regulator faces an absolute cost so high that abstention is dominant under the prior; even a high-bandwidth regulator may be priced out across the full product population. The condition that produces this state is not unique to AI. Pasquale (2015) documents the same inspection-cost architecture in algorithmic finance and consumer scoring: the opacity of the regulated object combines with the bandwidth of the regulator to make inspection a scarce resource allocated by triage rather than by uniform application.

Asynchronous harms. The user’s payoff term H_{θ_U} enters discounted. The episodes in which a relational reading proves costly—deprecation, persona shifts, crisis-handling failures—are temporally separated from the initial attribution by intervals over which the user has already accrued B_{θ_U} . Under standard hyperbolic or even exponential discounting, the inequality $B_{\theta_U} - \pi(L)H_{\theta_U} > 0$ holds for a wider class of users than the time-zero risk calculation would suggest. Asynchrony is not a behavioural bug; it is the timing structure of the harm itself.

Fragmentation of affected parties. Publics, journalists, intellectuals, and affected communities are folded into the noisy channel z rather than modelled as autonomous players. This is not a normative claim that they lack standing; it is an institutional observation. Their costs of coordination are high, their access to inspection technology is uneven, and their incentives to invest in any particular

case are individually small. Even when z carries substantial information about θ_F , regulators cannot treat it as binding without a procedural translation that the channel itself does not provide. The fragmentation result is closely related to the cheap-talk multiplicity of Crawford and Sobel (1982): with many would-be senders and a single noisy aggregator, the informative subset of equilibria is not selected for.

No penalty for inconsistency. The profitable cheap-talk cost structure sets $c_{\theta_F}(a) = c_{\theta_F}(p) = 0$ for both governance types and—more importantly for this section—imposes no joint constraint across m_F . Marketing copy and policy documentation are produced under different audiences, different review processes, and different liability regimes. Inconsistency between them is not detected as a single offence; it is decomposed into two compliant statements at two different points of contact. The firm’s premium Δ_F is rent earned precisely on this decomposition. Until the signal cost makes inconsistency itself sanctionable—the proposal of Section 8.1—there is no parameter under the firm’s control whose adjustment would close the spread.

Pooling is the stable outcome selected by the current incentive structure. These five structural conditions jointly explain why normative declarations do not change equilibrium behavior unless they alter the signal-cost structure. The constraint is not a failure of moral seriousness, but a formal property of the message space: a declaration of “responsible AI” is indistinguishable from any other message at $c_{\theta_F}(a) = 0$ while $\text{NetCost}_{\theta}(a) < 0$. Because both governance types can issue it at the same cost and capture the same premium, it contains no type information and moves no posterior. The pooling outcome is therefore the stable equilibrium of the incentive structure as currently constituted, even if it is welfare-reducing in aggregate. A firm that unilaterally forgoes the ontological premium Δ_F —that chooses deflationary marketing across all channels and forfeits the user investment it would otherwise attract—loses revenue it cannot recover from customers who face no such constraint. The implication is precise: within the pooling region defined above, unilateral non-arbitrage is not a best response; a firm that forgoes Δ_F loses the induced investment while competitors

retain it. The mechanisms in Section 8 respond to this implication by changing the cost parameters rather than appealing to the conscience of players who are already acting rationally.

Structural Condition	Model Parameter	Intervention (§8)
No penalty for inconsistency	$\text{NetCost}_\theta(a) < 0$	Sanctionable inconsistency
Opacity	θ_S private to firm	Costly third-party audits
Verification bandwidth	$K_{\theta_R} > \pi(L)D_R$	Auditable records
Asynchronous harms	H_{θ_U} enters discounted	Evidentiary lag priced
Fragmentation	z noisy, high coord. cost	Aggregation cost lowered

Table 2: Mapping of structural persistence conditions to model parameters and cost-imposition mechanisms.

7. Endogenous Trust Dynamics

The dynamic question is whether repeated interaction, user habituation, and delayed harms can substitute for externally priced signals. They cannot do so merely by adding history to the baseline game. The zero-retaliation boundary continues to matter dynamically because the anthropomorphic message expands the trust stock without imposing a corresponding production, verification, or sanction cost on the sender.

7.1. The Law of Motion for Endogenous Trust

Embed the baseline setting into an infinite-horizon Markov game. Let the public state variable $x_t \in X \subseteq \mathbb{R}_+$ denote endogenous trust capital, or relational reliance, accumulated in the user base at the beginning of period t . The state evolves according to

$$x_{t+1} = \alpha x_t + \beta i_t + \gamma \mathbf{1}_{\{m_t=a\}} - \delta q_t.$$

The state variable x_t is the aggregate stock of structural and affective reliance users place in the system. The firm’s stage-game ontological premium and

the user's baseline utility from investment are strictly increasing in x_t . The parameter $\alpha \in (0, 1)$ is trust persistence; $1 - \alpha$ is the natural depreciation of user trust in the absence of continued engagement or new trust-restoring messages. The action $i_t \in \{0, 1\}$ is user investment: $i_t = 1$ denotes investment, and $i_t = 0$ denotes detachment. The coefficient $\beta \geq 0$ is habituation or engagement reinforcement, so active user investment raises the next-period reliance baseline.

The indicator $\mathbf{1}_{\{m_t=a\}}$ equals one when the firm sends the trust-restoring anthropomorphic message and zero under deflationary positioning. The coefficient $\gamma > 0$ is the costless semantic premium: the deterministic expansion of trust capital generated by the anthropomorphic signal. Because the zero-retaliation boundary sets $c_\theta(a) = 0$, this state expansion is not matched by a production or verification cost. The shock $q_t \geq 0$ is asynchronous harm realization, drawn from $F(\cdot \mid \theta_F)$, with

$$\mathbb{E}[q_t \mid \theta_F = L] > \mathbb{E}[q_t \mid \theta_F = H].$$

It represents delayed leakage of private governance type through capability depreciation, unannounced persona shifts, abrupt refusals, or related adverse incidents. The parameter $\delta > 0$ is the depletion penalty per unit of asynchronous harm.

Theorem 3 (Almost-Sure Dynamic Pooling Invariance). *Let the endogenous trust state evolve according to*

$$x_{t+1} = \alpha x_t + \beta i_t + \gamma \mathbf{1}_{\{m_t=a\}} - \delta q_t,$$

where $\alpha \in (0, 1)$, $\beta \geq 0$, $\gamma > 0$, and $\delta > 0$. For each governance type $\theta \in \{H, L\}$, let

$$\text{supp}(q_t \mid \theta) \subseteq [0, \bar{q}_\theta],$$

with $\bar{q}_L > \bar{q}_H \geq 0$.

Let $I = [\underline{x}, \bar{x}] \subseteq \mathbb{R}_+$. Suppose that the semantic premium and habituation reinforcement satisfy

$$(1 - \alpha)\underline{x} + \delta \bar{q}_L \leq \beta + \gamma \leq (1 - \alpha)\bar{x}.$$

Suppose further that, for every $x \in I$, users strictly prefer investment after the anthropomorphic message under the pooling belief, and both firm types satisfy the Bellman incentive constraint for sending a rather than any alternative message.

Then there exists a Markov perfect pooling equilibrium on I in which both types send a , users invest, and the interval I is almost surely forward-invariant. That is, for every $x_t \in I$,

$$\mathbb{P}(\alpha x_t + \beta + \gamma - \delta q_t \in I \mid x_t \in I) = 1.$$

Consequently, along this equilibrium path, the trust state never escapes I despite recurring asynchronous harms.

Proof. Under the pooling profile, $m_t = a$ and $i_t = 1$, so

$$x_{t+1} = \alpha x_t + \beta + \gamma - \delta q_t.$$

For any $x_t \in I$ and any $q_t \in [0, \bar{q}_L]$, the transition is minimized at $x_t = \underline{x}$ and $q_t = \bar{q}_L$. Hence

$$x_{t+1} \geq \alpha \underline{x} + \beta + \gamma - \delta \bar{q}_L \geq \underline{x}$$

by the lower-bound condition. The transition is maximized at $x_t = \bar{x}$ and $q_t = 0$. Hence

$$x_{t+1} \leq \alpha \bar{x} + \beta + \gamma \leq \bar{x}$$

by the upper-bound condition. Therefore

$$x_{t+1} \in [\underline{x}, \bar{x}] = I$$

for every feasible shock realization. Since $\text{supp}(q_t \mid \theta) \subseteq [0, \bar{q}_\theta]$ and $\bar{q}_H < \bar{q}_L$, the same bound applies to both types. Thus I is forward-invariant almost surely.

The assumed user optimality and Bellman incentive constraints support the pooling Markov strategy profile on I . Therefore the invariant pooling path is a Markov perfect pooling equilibrium. $\square$

The logic is the dynamic counterpart of profitable cheap talk:

$$\gamma + \beta \quad \text{sustain the trust stock,}$$

$\Rightarrow$ pooling remains dynamically invariant.

The next issue is therefore not whether history exists, but whether history is priced. If a reputation or repair message remains costless at the zero-retaliation boundary, it is another trust-restoring signal available to the low-governance type. Dynamic separation requires a priced wedge: audit, certification, sanctionable inconsistency, or another mechanism that changes the Bellman value comparison rather than merely adding an unpriced history variable.

7.2. Dynamic Shadowing and No Trust-Restoring History

The dynamic environment reverses the Spence logic. In a standard costly-signaling model, the high type values separation more because it can bear the signal cost more cheaply. Here, at the zero-retaliation boundary, the signal is unpriced. Since the low-governance type loses trust faster on the pooling path, a free trust-restoring message creates at least as large a net continuation gain for the low type as for the high type. The deviation is therefore shadowed rather than separating.

Theorem 4 (Dynamic Shadowing and Endogenous Reverse Single-Crossing). *Let $\bar{V}_\theta(x_t)$ denote the continuation value of type $\theta \in \{H, L\}$ on the almost-sure pooling path at state $x_t \in I$. Suppose a high-governance firm attempts to break the pooling equilibrium by issuing an unpriced, history-dependent trust-restoring neologism $c(h_t)$. Let $V_\theta(c, x_t)$ denote the continuation value of type θ if receivers accept the message as credible.*

A dynamically credible separating deviation requires

$$V_H(c, x_t) - \bar{V}_H(x_t) > 0$$

and

$$V_L(c, x_t) - \bar{V}_L(x_t) \leq 0.$$

At the zero-retaliation boundary, the message $c(h_t)$ imposes no type-differential execution cost. Moreover, because the low-governance type faces stochastically

larger asynchronous harms on the pooling path, its pooling continuation value is weakly lower:

$$\bar{V}_L(x_t) \leq \bar{V}_H(x_t).$$

Assume further that the accepted neologism weakly compresses the type-continuation wedge:

$$V_H(c, x_t) - V_L(c, x_t) \leq \bar{V}_H(x_t) - \bar{V}_L(x_t).$$

Then the net continuation premium satisfies the reverse single-crossing inequality

$$V_L(c, x_t) - \bar{V}_L(x_t) \geq V_H(c, x_t) - \bar{V}_H(x_t).$$

Consequently, if the neologism is strictly profitable for the high type,

$$V_H(c, x_t) - \bar{V}_H(x_t) > 0,$$

then it is also strictly profitable for the low type:

$$V_L(c, x_t) - \bar{V}_L(x_t) > 0.$$

Thus every profitable unpriced trust-restoring neologism is dynamically shadowed by the low-governance type and cannot sustain type separation.

Proof. The wedge-compression condition gives

$$V_H(c, x_t) - V_L(c, x_t) \leq \bar{V}_H(x_t) - \bar{V}_L(x_t).$$

Rearranging yields

$$V_L(c, x_t) - \bar{V}_L(x_t) \geq V_H(c, x_t) - \bar{V}_H(x_t).$$

Hence, whenever

$$V_H(c, x_t) - \bar{V}_H(x_t) > 0,$$

we also have

$$V_L(c, x_t) - \bar{V}_L(x_t) > 0.$$

The low-governance type therefore strictly benefits from issuing the same history-dependent neologism. The deviation cannot be assigned exclusively to the high-governance type under any credible separating belief. Therefore the unpriced trust-restoring message is dynamically shadowed. $\square$

Corollary 3 (Symmetric Trust Reset). *If an accepted trust-restoring neologism resets the receiver-facing trust state symmetrically so that*

$$V_L(c, x_t) = V_H(c, x_t),$$

then the wedge-compression condition holds automatically whenever

$$\bar{V}_L(x_t) \leq \bar{V}_H(x_t).$$

Thus every profitable high-type neologism is strictly shadowed by the low type.

Proof. If $V_L(c, x_t) = V_H(c, x_t)$, then

$$V_H(c, x_t) - V_L(c, x_t) = 0.$$

The condition $\bar{V}_L(x_t) \leq \bar{V}_H(x_t)$ gives

$$0 \leq \bar{V}_H(x_t) - \bar{V}_L(x_t),$$

which is exactly the wedge-compression condition. The shadowing conclusion then follows from Theorem 4. $\square$

7.3. Externally Priced Dynamic Separation

The dynamic Spence ratio shows why the standard costly-signaling intuition is incomplete in endogenous trust environments. It is not enough that the low-governance type faces a higher marginal audit cost. Because the low type also receives a larger dynamic premium from escaping the deteriorating pooling path, the audit technology must discriminate faster than the endogenous continuation premium grows. Separation requires not merely $c_L > c_H$, but

$$\frac{c_L}{c_H} > \frac{\Delta V_L}{\Delta V_H}.$$

Theorem 5 (Externally Priced Separation via Dynamic Single-Crossing). *Consider the almost-sure pooling regime supported on the invariant trust interval I . Let an external mechanism make available a verifiable signal s with type- and state-dependent cost $k_\theta(s, x)$, where $\theta \in \{H, L\}$.*

Let $V_\theta^S(x)$ denote the before-cost expected continuation value of successfully separating and securing the relational premium, and let $V_\theta^P(x)$ denote the continuation value of remaining on the unverified pooling path. Define

$$\Delta V_\theta(x) := V_\theta^S(x) - V_\theta^P(x).$$

Within the class of audit-supported Markov equilibria, the signal s supports a separating Markov perfect equilibrium on I if and only if, for every $x \in I$,

$$k_L(s, x) > \Delta V_L(x)$$

and

$$k_H(s, x) \leq \Delta V_H(x).$$

The first inequality deters the low-governance type from mimicking the separating signal; the second gives the high-governance type a weakly profitable incentive to separate.

Proof. At state x , the low-governance type's net gain from mimicking the separating signal is

$$\Delta V_L(x) - k_L(s, x).$$

The low type is deterred precisely when this quantity is strictly negative, equivalently $k_L(s, x) > \Delta V_L(x)$. The high-governance type's net gain from sending the separating signal is

$$\Delta V_H(x) - k_H(s, x).$$

The high type weakly prefers separation precisely when this quantity is non-negative, equivalently $k_H(s, x) \leq \Delta V_H(x)$. These two incentive constraints are necessary for a separating Markov equilibrium supported by the audit signal, and, within the stated class of audit-supported Markov equilibria, they are sufficient: assign s to the high type, assign the unverified pooling action to the low type, and let receivers treat the verified signal as credible. $\square$

Remark 2 (Structural wedge). Recall from Theorem 4 that unpriced semantic claims yield an endogenous reverse single-crossing: $\Delta V_L(x) \geq \Delta V_H(x)$. A

necessary implication of the two inequalities in Theorem 5 is

$$k_L(s, x) - k_H(s, x) > \Delta V_L(x) - \Delta V_H(x) \geq 0.$$

Thus the external mechanism must generate a cost wedge large enough to overpower the low type's endogenous trust-reset advantage.

Corollary 4 (Separating Audit Intensity and the Dynamic Spence Ratio). *Suppose the external mechanism is parameterized by a scalar audit intensity $e \geq 0$, with linear type-dependent costs*

$$k_\theta(e, x) = c_\theta(x)e,$$

where

$$c_L(x) > c_H(x) > 0.$$

Assume further that

$$\Delta V_L(x) \geq \Delta V_H(x) > 0.$$

A state-contingent audit intensity $e^*(x)$ supporting separation at state x exists if and only if

$$\frac{c_L(x)}{c_H(x)} > \frac{\Delta V_L(x)}{\Delta V_H(x)} \geq 1.$$

When this ratio condition holds, any audit intensity satisfying

$$e^*(x) \in \left(\frac{\Delta V_L(x)}{c_L(x)}, \frac{\Delta V_H(x)}{c_H(x)} \right]$$

supports separation at state x . If the ratio condition fails, the separating interval is empty, and every audit intensity affordable to the high-governance type is also profitably shadowed by the low-governance type.

Proof. By Theorem 5, separation at state x requires

$$c_L(x)e > \Delta V_L(x) \quad \text{and} \quad c_H(x)e \leq \Delta V_H(x).$$

Since $c_L(x) > c_H(x) > 0$, this is equivalent to

$$e > \frac{\Delta V_L(x)}{c_L(x)} \quad \text{and} \quad e \leq \frac{\Delta V_H(x)}{c_H(x)}.$$

The feasible interval is nonempty if and only if

$$\frac{\Delta V_L(x)}{c_L(x)} < \frac{\Delta V_H(x)}{c_H(x)},$$

which, because $\Delta V_H(x) > 0$ and $c_H(x) > 0$, is equivalent to

$$\frac{c_L(x)}{c_H(x)} > \frac{\Delta V_L(x)}{\Delta V_H(x)}.$$

The lower bound $\Delta V_L(x)/\Delta V_H(x) \geq 1$ follows from the assumed reverse dynamic premium. $\square$

Corollary 5 (Uniform Audit Intensity). *A single audit intensity e supports separation throughout the interval I if and only if*

$$\sup_{x \in I} \frac{\Delta V_L(x)}{c_L(x)} < \inf_{x \in I} \frac{\Delta V_H(x)}{c_H(x)}.$$

In that case, any

$$e \in \left(\sup_{x \in I} \frac{\Delta V_L(x)}{c_L(x)}, \inf_{x \in I} \frac{\Delta V_H(x)}{c_H(x)} \right]$$

supports a separating Markov equilibrium on I .

Proof. For a fixed e to support separation at every $x \in I$, it must satisfy

$$e > \frac{\Delta V_L(x)}{c_L(x)} \quad \text{and} \quad e \leq \frac{\Delta V_H(x)}{c_H(x)}$$

for every $x \in I$. This is equivalent to

$$e > \sup_{x \in I} \frac{\Delta V_L(x)}{c_L(x)} \quad \text{and} \quad e \leq \inf_{x \in I} \frac{\Delta V_H(x)}{c_H(x)}.$$

Such an e exists if and only if the displayed strict inequality between the supremum and infimum holds. $\square$

This corollary separates pointwise mechanism design from institutional design:

$$\text{local audit feasibility} \neq \text{uniform institutional feasibility}.$$

An audit technology can separate types at each isolated state while still failing to provide a single administrable intensity that works across the entire trust interval.

8. Pricing the Ontological Spread: Necessity and Sufficiency Conditions

The three interventions below are an external pricing architecture. The section moves from formal model to mechanism implication to institutional implementation:

formal model $\rightarrow$ mechanism implication $\rightarrow$ institutional implementation.

Sanctionable inconsistency prices the cross-channel spread. Audits price the anthropomorphic claim. Auditable records price verification by changing the regulator's inspection inequality. The mechanisms therefore operate on different margins of the same zero-retaliation arbitrage condition.

Definition 1 (Low-type arbitrage spread). Define the low-type arbitrage spread from a trust-restoring register q as

$$\Omega_L(q) := G(q) - C_L(q) - \Pi_L(q),$$

where $G(q)$ is the gross continuation gain from the register, $C_L(q)$ is the low type's signal or compliance cost, and $\Pi_L(q)$ is the expected enforcement price attached to that register. A low-governance type profitably shadows whenever $\Omega_L(q) > 0$. A mechanism disciplines profitable cheap talk only if it drives $\Omega_L(q) \leq 0$ while preserving $\Omega_H(q) \geq 0$ for the desired high-governance signal.

Proposition 2 (Pricing the ontological spread). *Let q be a trust-restoring register or claim. In the profitable cheap-talk baseline, $C_H(q) = C_L(q) = 0$ and $\Pi_H(q) = \Pi_L(q) = 0$, so*

$$\Omega_H(q) = \Omega_L(q) = G(q) > 0.$$

Thus q cannot separate types. A mechanism can restore separation or prune arbitrage only by altering at least one of three terms:

$$\Omega_\theta(q) = G(q) - C_\theta(q) - \Pi_\theta(q).$$

- (i) A sanctionable-inconsistency rule raises $\Pi_L(q)$ for inconsistent cross-channel registers.
- (ii) An audit requirement raises $C_\theta(q)$ in a type-dependent way.
- (iii) Auditable records raise the probability or payoff of enforcement by lowering K_R , increasing D_R , or increasing the detection probability embedded in $\Pi_\theta(q)$.

For a separating trust-restoring signal to exist, it is necessary that some mechanism create a wedge such that

$$\Omega_H(q) \geq 0 \quad \text{and} \quad \Omega_L(q) \leq 0.$$

Absent such a wedge, every trust-restoring message remains shadowable. Within the audit family $c_\theta(e, \theta_S) = k_\theta e + \theta_S$, the single-crossing condition

$$c_H(e, \theta_S) \leq G(q) \leq c_L(e, \theta_S)$$

is necessary and sufficient for separation by audit intensity.

Remark 3 (Scope of the exhaustiveness claim). *The claim that a mechanism can operate only through G , C_θ , or Π_θ is exhaustive within the single-shot baseline of Section 3, where payoffs are stage-game quantities and $\Omega_\theta(q)$ is defined over a single interaction. In dynamic or multi-product extensions, reputation effects may lower the effective premium G through adverse posteriors that persist across periods, and multi-product cost spillovers may raise C_θ as a by-product of compliance investments elsewhere in the firm's portfolio. Either channel could discipline the pooling equilibrium without an explicit external instrument. Those extensions are consistent with the pricing proposition; they instantiate it by altering G and C_θ through endogenous market forces rather than regulatory mandates. The present paper establishes conditions under which such endogenous forces are absent and external cost-imposition is necessary—the zero-retaliation, no-reputation, single-product limit studied in Sections 2–6.*

Corollary 6 (Sufficiency of sanctionable inconsistency for register pruning). *For an inconsistent cross-channel register $q = (a, p)$, a penalty for sanctionable*

inconsistency prunes the register whenever

$$\rho_R SI \geq \Delta_F^S.$$

If $\rho_R SI = 0$ and no other mechanism prices inconsistency, then q remains profitable whenever $\Delta_F^S > 0$.

Corollary 7 (Necessary and sufficient audit condition). *Within the audit family $c_\theta(e, \theta_S) = k_\theta e + \theta_S$ with $0 < k_H < k_L$, an audited anthropomorphic signal supports separation if and only if there exists $e \geq 0$ such that*

$$c_H(e, \theta_S) \leq G \leq c_L(e, \theta_S).$$

Equivalently, when $G > \theta_S$,

$$e \in \left[\frac{G - \theta_S}{k_L}, \frac{G - \theta_S}{k_H} \right].$$

Corollary 8 (Verification feasibility). *Auditable records make active regulation sequentially rational whenever*

$$K_R - \lambda \leq \pi(L)(D_R + \eta).$$

If this inequality fails, recordkeeping alone does not discipline profitable cheap talk; it stores information without making enforcement incentive-compatible.

Complementarity. The mechanisms are not substitutes. They price different margins. Sanctionable inconsistency without records imposes a penalty that the regulator cannot verify—a paper tiger when ρ_R is low. Records without sanctionable inconsistency store information without attaching a price: an archive, not a mechanism. Audit requirements without sanctionable inconsistency can support a verified (a, a) claim, but cannot automatically prohibit the cross-channel (a, p) arbitrage register. Sanctionable inconsistency combined with records prunes inconsistent registers. Audit requirements combined with records support credible high-governance agentic claims. All three together convert profitable cheap talk into costly signaling with enforceable consistency: the

Mechanism	Prices what?	Sufficient for	Necessary when
Sanctionable inconsistency	cross-channel spread (a, p)	pruning inconsistent register if $\rho_R SI \geq \Delta_F^S$	no other mechanism prices inconsistency
Costly audit	anthropomorphic claim a	type separation $c_H \leq G \leq c_L$	if separation by message requires type-dependent cost wedge
Auditable records	verification / enforcement feasibility	inspection if $K_R - \lambda \leq \pi(\mathbf{L})(D_R + \eta)$	detection probability or verification payoff remains too low

Table 3: Mechanism matrix: each intervention prices a different margin of the arbitrage spread $\Omega_\theta(q)$.

Mechanism	Model term priced	Institutional form
Sanctionable inconsistency	(a, p) register spread	net-impression / substantiation review
Costly audit	$k_\theta(s, x)$	substantiation-before-reliance
Auditable records	K_R, D_R	record-backed oversight

Table 4: Institutional forms of external pricing.

arbitrage spread $\Omega_L(q)$ is attacked on the cost term by audits, on the penalty term by sanctions, and on the enforcement-feasibility precondition by records.

The subsections below elaborate each mechanism. Each states the formal intervention on model parameters, the equilibrium shift, the necessity and sufficiency conditions, and the institutional form through which the relevant model term is externally priced.

8.1. Sanctionable Inconsistency

The first intervention treats the firm's public message as a joint object rather than as two isolated texts. In the profitable cheap-talk model, marketing can send a while policy and liability documents send p , with no cost assigned to the inconsistency.

The consumer-side informational asymmetry is concrete. A user reads “your AI companion” on a product page, encounters warmth designed to induce investment, and is later told—in support documentation the user will never read—that no relational claim was ever made. Nothing in the user’s information set indicates that the two artefacts were authored by the same actor for different audiences. The inconsistency is costless because no single document is false; each is a compliant statement at a different point of contact.

A sanctionable-inconsistency rule would require firms to maintain a claim register $\mathcal{R} := C \rightarrow M_F$ linking outward-facing marketing, capability statements, terms of service, incident disclosures, and liability positions, where the baseline firm message space remains $M_F = \{a, p\}$. If the firm represents the system as agentic, caring, autonomous, or safety-critical in one channel while denying the legal significance of the same representation in another, that cross-channel pair (a, p) becomes independently sanctionable as an inconsistency in $\mathcal{R}$.

Formal Intervention.. In the notation of Proposition 2, sanctionable inconsistency raises the penalty term $\Pi_L(q)$ by adding S_I to the firm’s payoff for inconsistent message pairs. If ρ_R is the probability that R detects or accepts proof of the inconsistency, the profitable cheap-talk premium becomes

$$\Delta_F - \rho_R S_I.$$

Pooling on anthropomorphic marketing is no longer profitable when $\rho_R S_I > \Delta_F$. The rule therefore attacks the exact term that made arbitrage attractive: the rent earned from pricing the system as agent in the user channel while pricing it as tool in the liability channel.

Necessity and Sufficiency.. Sanctionable inconsistency is not a separation mechanism. It is a register-pruning mechanism: its function is to delete the inconsistent register $q = (a, p)$ from the profitable message space, not to distinguish high-governance from low-governance types. For an inconsistent register $q = (a, p)$, sanctionable inconsistency is sufficient to prune q whenever

$$\rho_R S_I \geq \Delta_F^S - [c_\theta(a, p) - c_\theta(p, p)].$$

In the baseline $c_\theta(a, p) = c_\theta(p, p) = 0$, this reduces to $\rho_R SI \geq \Delta_F^S$ (Corollary 6). If inconsistency remains unpriced, i.e. $\rho_R SI = 0$, and no audit or recordkeeping mechanism changes the expected payoff of (a, p) , then the inconsistent register remains profitable whenever $\Delta_F^S > 0$. Thus some positive price on inconsistency is necessary to eliminate pure cross-channel arbitrage in the baseline.

Equilibrium Shift.. The induced shift is not necessarily full separation. It can also operate by constraining off-path beliefs. If R observes an inconsistent pair (a, p) , then the admissible posterior places greater mass on $\theta_F = \mathbf{L}$ or on high opacity θ_S . Firms anticipating that posterior either harmonize their claims or choose the deflationary message across channels. In both cases, the profitable cheap-talk pooling equilibrium loses its most valuable deviation: firms can no longer take the marketing upside of a while preserving the liability downside of p . The Isabelle/HOL appendix mechanizes the corresponding pruning claims: under the stated parameter conditions, neither inconsistent nor all-agentic registers can be supported in any Perfect Bayesian Equilibrium.

8.2. Costly Signaling Through Third-Party Audits

The second mechanism makes anthropomorphic or agency-adjacent marketing a costly signal. A firm may still describe a system as a companion, agent, co-worker, tutor, or high-reliance assistant, but only after a qualified third party verifies the governance conditions that make such a claim non-misleading: evaluation coverage, red-team results, escalation pathways, incident-response capacity, model-update controls, and post-deployment monitoring. The audit does not certify consciousness or moral status. It certifies the institutional conditions behind the claim.

Formal Intervention.. This implements the audit-trail cost from Theorem 2. Let audit intensity be e , with $c_\theta(e, \theta_S) = k_\theta e + \theta_S$ for type $\theta \in \{\mathbf{H}, \mathbf{L}\}$, where $0 < k_H < k_L$. The separating interval derived in Section 5 is

$$\frac{G - \theta_S}{k_L} \leq e \leq \frac{G - \theta_S}{k_H},$$

provided $G > \theta_S$. Equivalently, the incentive constraints are

$$c_H(e, \theta_S) \leq G \leq c_L(e, \theta_S).$$

Above the lower threshold, low-governance firms cannot profitably mimic. Below the upper threshold, high-governance firms can still afford to signal. The regulator’s task is therefore not to maximize audit burden. It is to set e in the interval where signal cost separates types.

Necessity and Sufficiency. Costly audit signals are the type-separation mechanism proper. Within the audit-augmented message space, an audit intensity e supports separation if and only if

$$c_H(e, \theta_S) \leq G \leq c_L(e, \theta_S)$$

(Corollary 7). The regulator’s task is not to maximize audit burden. It is to locate the single-crossing interval. In the absence of any type-dependent cost wedge—that is, if $k_H = k_L$ —no audit-like message can be type-exclusive; it collapses back into profitable cheap talk. The single-crossing condition $k_H < k_L$ is therefore necessary, not merely sufficient, for audit-based separation.

Equilibrium Shift. The shift is from pooling to separating. High-governance firms send audited anthropomorphic claims; low-governance firms either invest in governance until their cost curve changes or use deflationary marketing because mimicry is no longer privately profitable. Users can then condition investment on an informative signal rather than on undisciplined warmth in product copy. Regulators can condition abstention on the audit result rather than on an unverified prior. This is the Spence logic of costly signaling applied to AI governance claims (Spence, 1973); it also follows the advertising-as-signal insight that market messages become informative only when false messages are sufficiently expensive (Milgrom and Roberts, 1986). The Isabelle/HOL appendix mechanizes the corresponding opacity condition: when system opacity is high enough, even a high-governance firm cannot profitably send the audited signal.

8.3. Auditable Records for Regulator-Readable Verification

The third intervention reduces the regulator’s verification cost by requiring append-only, regulator-readable records tied to capability and relational claims. Records should include the claim identifier, model version, relevant evaluation suite, deployment date, material model updates, risk controls, known incident classes, and the internal owner responsible for the claim. The point is not to publish every interaction or expose user data. It is to create a durable evidentiary substrate that lets R test whether a public claim remained true across model updates and incidents.

Formal Intervention.. In the pooling condition from Section 3, the regulator abstains when $K_{\theta_R} > \pi(L)D_R$. In the notation of Proposition 2, auditable records operate on the enforcement-feasibility precondition embedded in $\Pi_\theta(q)$ by lowering K_{θ_R} and increasing expected detection value D_R :

$$K'_{\theta_R} = K_{\theta_R} - \lambda, \quad D'_R = D_R + \eta.$$

When $K'_{\theta_R} \leq \pi(L)D'_R$, inspection becomes sequentially rational (Corollary 8).

Necessity and Sufficiency.. Auditable records are not themselves a separation or pruning mechanism. They are an enforcement-feasibility condition: they make sanctions and audits administrable by transforming the regulator’s inspection inequality from $K_R > \pi(L)D_R$ to $K_R - \lambda \leq \pi(L)(D_R + \eta)$. Records are sufficient to flip the regulator’s best response to inspection when the transformed inequality holds, but they are not sufficient by themselves to price the arbitrage spread unless linked to a penalty or claim obligation. If $K_R > \pi(L)D_R$ remains true after the intervention, then recordkeeping alone does not discipline the pooling equilibrium. It may store information, but it does not make verification incentive-compatible. Records without inspection incentives are archives, not mechanisms.

Equilibrium Shift.. The shift is from abstention under unverifiable priors to credible inspection. Once firms know that claims are tied to records, a low-governance firm faces a higher expected cost from exaggeration even if it is not

inspected in every case. This complements the first two proposals. Sanctionable inconsistency tells firms which cross-channel contradictions will be penalized; costly signaling tells firms what evidence is required before making anthropomorphic claims; auditable records make both rules administrable over time. The Isabelle/HOL appendix mechanizes this best-response flip from abstention to active intervention under the record-keeping parameter condition.

8.4. *Positioning*

The baseline communication problem is profitable cheap talk. Crawford and Sobel (1982) show that a sender with private information can transmit only coarse information when messages are free; the present model adds the zero-retaliation boundary at which the message is not only free, but payoff-positive to send. The separation result follows the costly-signal logic of Spence (1973) and Milgrom and Roberts (1986): a message becomes informative only when false or low-quality senders face higher marginal cost. The refinement issue is closer to Farrell’s neologism-proofness (Farrell, 1993) than to standard type-exclusion refinements (Cho and Kreps, 1987; Banks and Sobel, 1987), because the disputed message has a literal trust-restoring meaning. At $\rho_S = 0$, that literal meaning cannot be made credible without a cost that excludes the low-governance type.

This implication is narrower than principle-based AI ethics (Floridi et al., 2018; Jobin et al., 2019). Voluntary principles are cheap-talk messages unless they are tied to audits, records, or sanctions. Algorithmic-accountability proposals identify the opacity problem (Pasquale, 2015); the model here states the single-crossing condition under which transparency instruments become type-separating. The Isabelle/HOL development is treated as an audit artifact rather than the central contribution: it verifies the finite equilibrium constructions, while the game-theoretic contribution is the payoff-relevant non-player and the zero-retaliation shadowing result.

9. Conclusion

This paper identifies a prior obstruction to common-knowledge formation. When the trust-restoring messages by which posteriors would become common knowledge are themselves profitable to distort, rational senders need not converge to truthful agreement. They converge to arbitrage-compatible pooling. Anthropomorphic AI marketing is the leading application because the classified object is payoff-relevant but cannot impose retaliation costs on the classification.

The equilibrium is not a moral failure. The pooling Perfect Bayesian Equilibrium documented in Section 3 does not require bad actors. It requires only that the anthropomorphic signal have negative net cost, that immediate user benefit exceed discounted expected harm, and that regulator inspection cost exceed expected damage under the prior. Under those conditions, every player is already acting rationally. As Section 6 shows, moral seriousness is itself a profitable cheap-talk signal unless it is tied to audit, record, or sanction costs. Firms that do not capture recognition rent are outcompeted by firms that do.

The payoff-relevant non-player creates profitable cheap talk. The subject-retaliation parameter ρ_S captures a structural fact about prior recognition games: the classified subject’s non-zero retaliation capacity bounded the premium Δ_F^S from above. The AI case eliminates this capacity entirely. The result is that $\Delta_F^S = \Delta_F$ and $\text{NetCost}_\theta(a) = -\Delta_F < 0$: the system can be the object of strategic classification without being a player capable of disciplining that classification.

The mechanism is cost imposition, not dignity. Sanctionable inconsistency, third-party audit requirements, and auditable records are interventions that transform profitable cheap talk into costly signaling. Their equilibrium effect is to make it expensive for firms to describe AI systems as companions, agents, or safety-critical partners without substantiation. The Isabelle/HOL appendix mechanizes the same pruning logic: inconsistent and all-agentic registers are excluded under their respective parameter conditions, while audited agentic claims survive only when verification satisfies the audit condition. Regulation

achieves separation not by presuming voluntary disclosure, but by changing the payoff structure that determines what firms rationally say.

References

- Akerlof, G.A., 1970. The market for “lemons”: Quality uncertainty and the market mechanism. *The Quarterly Journal of Economics* 84, 488–500. doi:10.2307/1879431.
- Aumann, R.J., 1976. Agreeing to disagree. *The Annals of Statistics* 4, 1236–1239. doi:10.1214/aos/1176343654.
- Banks, J.S., Sobel, J., 1987. Equilibrium selection in signaling games. *Econometrica: Journal of the Econometric Society* 55, 647–661.
- Cho, I.K., Kreps, D.M., 1987. Signaling games and stable equilibria. *The Quarterly Journal of Economics* 102, 179–221. doi:10.2307/1885060.
- Crawford, V.P., Sobel, J., 1982. Strategic information transmission. *Econometrica* 50, 1431–1451. doi:10.2307/1913390.
- Farrell, J., 1993. Meaning and credibility in cheap-talk games. *Games and Economic Behavior* 5, 514–531. doi:10.1006/game.1993.1029.
- Floridi, L., Cowls, J., Beltrametti, M., Chatila, R., Chazerand, P., Dignum, V., Luetge, C., Madelin, R., Pagallo, U., Rossi, F., Schafer, B., Valcke, P., Vayena, E., 2018. AI4People—an ethical framework for a good AI society: Opportunities, risks, principles, and recommendations. *Minds and Machines* 28, 689–707. doi:10.1007/s11023-018-9482-5.
- Jobin, A., Ienca, M., Vayena, E., 2019. The global landscape of AI ethics guidelines. *Nature Machine Intelligence* 1, 389–399. doi:10.1038/s42256-019-0088-2.
- Kreps, D.M., Wilson, R., 1982. Sequential equilibria. *Econometrica* 50, 863–894. doi:10.2307/1912767.

- Milgrom, P., Roberts, J., 1986. Relying on the information of interested parties. *The RAND Journal of Economics* 17, 18–32. doi:10.2307/2555625.
- Pasquale, F., 2015. *The Black Box Society: The Secret Algorithms That Control Money and Information*. Harvard University Press, Cambridge, MA.
- Spence, M., 1973. Job market signaling. *The Quarterly Journal of Economics* 87, 355–374. doi:10.2307/1882010.